

# Youngjae Cho

Machine Learning Researcher

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youngjae-cho.github.io | GitHub | Google Scholar

I study **robust and data-efficient machine learning**, with a focus on preference optimization, active learning, Bayesian adaptation, and video understanding.

## EDUCATION

**KAIST | M.S., Industrial & Systems Engineering**

Mar 2022 - Feb 2024

Advisor: Il-Chul Moon

**KAIST | B.S., Industrial & Systems Engineering**

Mar 2017 - Feb 2022

## SELECTED PUBLICATIONS

**NeurIPS 2026**

Accepted

First author

**Learning Where It Matters: Geometric Anchoring for Robust Preference Alignment**

**Youngjae Cho**, Jongsuk Kim, Ji-Hoon Kim

Geometric anchoring for robust preference optimization; **+3.6 percentage points** in AlpacaEval 2.0 LC win rate over SimPO.

**arXiv 2026**

Preprint

First author

**Beyond Scalar IoU: Structured Verification from Rollout Groups for Video Temporal Grounding**

**Youngjae Cho**, Won Young Jhoo, Jongsuk Kim

**SUTURE**: Rollout-group verification for video temporal grounding, improving performance across five benchmarks.

**AAAI 2024**

First author

**Make Prompts Adaptable: Bayesian Modeling for Vision-Language Prompt Learning with Data-Dependent Prior**

**Youngjae Cho**, HeeSun Bae, Seungjae Shin, YeoDong Youn, Weonyoung Joo, Il-Chul Moon

Bayesian prompt adaptation under scarce data and distribution shift.

**ICML 2023**

Co-first author

**Sharpness-Aware Active Learning**

Yoon-Yeong Kim\*, **Youngjae Cho**\*, JoonHo Jang, Byeonghu Na, Yeongmin Kim, Kyungwoo Song, Wanmo Kang, Il-Chul Moon

Sharpness-aware acquisition for label-efficient learning.

\* Equal contribution. Additional publications appear on page 2.

RESEARCH EXPERIENCE

Pyler

Research Scientist

Oct 2025 - Present

**Preference optimization.** Developed GAPO (NeurIPS 2026, accepted) and multimodal preference-optimization methods for Nemotron-Nano-12B-v2-VL using SimPO and Megatron-Bridge.

**Video understanding.** Developed SUTURE, a rollout-group verifier for video temporal grounding (arXiv 2026; first author).

**Auditable supervision.** Recovered decision trees from 299K LLM/VLM reasoning traces; improved content-safety macro F1 from **0.777 to 0.857**.

Aiv Co.

ML Research Scientist

Mar 2024 - Oct 2025

**Generative modeling.** Led background-aware diffusion research for industrial anomaly detection. Evaluated on MVTec-AD and LOCO and deployed defect synthesis to improve production detector precision and recall.

KAIST

Graduate Research

Mar 2022 - Feb 2024

**Data-efficient learning.** Developed Bayesian vision-language prompt adaptation (APP, AAAI 2024; first author) and sharpness-aware active learning (SAAL, ICML 2023; co-first author).

ADDITIONAL PUBLICATIONS

|                                        |                                                                                                                                                                                                                                                                                    |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| arXiv 2024<br>Preprint<br>First author | <b>Background-Aware Defect Generation for Robust Industrial Anomaly Detection</b><br>Youngjae Cho, Gwangyeol Kim, Sirojbek Safarov, Seongdeok Bang, Jaewoo Park                                                                                                                    |
| ICML 2022 Workshop                     | <b>Improving Group-based Robustness and Calibration via Ordered Risk and Confidence Regularization</b><br>Seungjae Shin, Byeonghu Na, HeeSun Bae, JoonHo Jang, Hyemi Kim, Kyungwoo Song, Youngjae Cho, Il-Chul Moon<br>Workshop on Spurious Correlations, Invariance and Stability |
| IEEE SMC 2021                          | <b>Predict Sequential Credit Card Delinquency with VaDE-Seq2Seq</b><br>Yeongmin Kim, Youngjae Cho, Hanbit Lee, Il-Chul Moon                                                                                                                                                        |

ACADEMIC SERVICE

ICML 2026 | Gold Reviewer

NeurIPS 2026 | Reviewer

AWARDS

NVIDIA Nemotron Hackathon

Track B Winner | 2026

Team winner, Domain-Specialized Model track.